

Shifting, Reflecting, and Stretching Quadratic Graphs

Vertex

Axis of Symmetry

Standard Form: $y = ax^2 + bx + c$

$$\left(\frac{-b}{2a}, f\left(\frac{-b}{2a}\right) \right)$$

$$x = \frac{-b}{2a}$$

Vertex Form: $y = a(x-h)^2 + k$

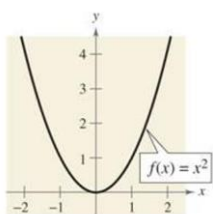
$$(h, k)$$

$$x = h$$

Parent Graph

Vertical & Horizontal Transformations

(does NOT change the shape of the graph, just shifts it)



- Let c be a positive real number.

- $h(x) = f(x) + c$

- If $c > 0$, the graph will shift UP c units

- If $c < 0$, the graph will shift DOWN c units

- $h(x) = f(x - c)$

- If $c > 0$, the graph will shift RIGHT c units

- If $c < 0$, the graph will shift LEFT c units

**The numbers inside the () shifts the graph left/right.

**The numbers added/subtracted outside the () shifts the graph

**Using a graphing calculator, graph the following:

$$Y_1 = x^2$$

$$Y_2 = (x - 4)^2$$

$$Y_3 = (x + 4)^2$$

$$Y_4 = x^2 + 4$$

$$Y_5 = x^2 - 4$$

Example 1: Describe the transformation of the following functions:

Example	Transformation
1. $f(x) = x^2 + 3$	
2. $f(x) = (x - 5)^2$	
3. $f(x) = x^2 - 2$	
4. $f(x) = (x + 2)^2 - 3$	
5. $f(x) = (x - 1)^2 - 4$	

Reflective Transformations (does NOT change the shape of the graph, just reflects it)

- Let c be a positive Real number:
 - $h(x) = -f(x)$ The graph is reflected across the x – axis (all the positive y – values are now negative, and vice versa)
 - $h(x) = f(-x)$ The graph is reflected across the y – axis (when you plug in negative x – values, you get the same y – values as when you plug in positive x 's)

****Using a graphing calculator, graph the following:**

$$Y_1 = (x - 2)^2$$

$$Y_2 = -(x - 2)^2 = -Y_1$$

$$Y_3 = (-x - 2)^2$$

Example 2: Describe the transformation of the following functions:

Example	Transformation
1. $f(x) = -x^2 - 5$	
2. $f(x) = (-x + 2)^2$	
3. $f(x) = (-x)^2 + 4$	
4. $f(x) = -(-x + 2)^2$	

Non-Rigid Transformations (changes the shape of the graph by shrinking or stretching the graph)

- Let c be a positive Real number:

- $g(x) = c \cdot f(x)$

- If $c > 1$, the graph is stretched vertically
 - If $0 < c < 1$, the graph will shrink

Note: Since we define c to be a positive value, $c > 0$. Why do we not consider when $c = 1$ (because it would not shrink nor stretch any!)

****Using a graphing calculator, graph the following:**

$$Y_1 = x^2$$

$$Y_2 = 5x^2$$

$$Y_3 = \frac{1}{5}x^2$$

Example 3: Determine the vertex and axis of symmetry.

A) $f(x) = -(-x+2)^2$

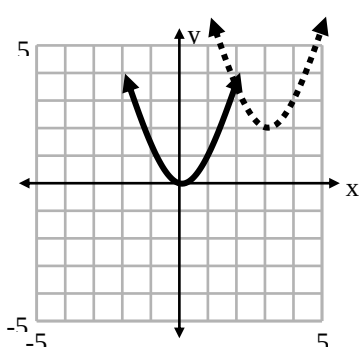
B) $f(x) = -x^2 - 5$

C) $f(x) = 2(x-1)^2 - 4$

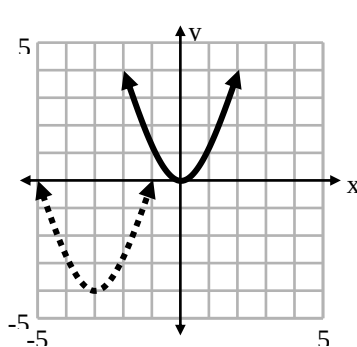
D) $f(x) = (x+2)^2 - 3$

Example 4: Using transformations, write equation for each function (in dashes):

A)



B)



C)

